

Machima2 Multiplicative Update Rules

symmetric epigenome variant, list mode

AIRE-experiments / Machima v1.1.0

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1 Setup and notation

Inputs (per chromosome $k = 1, \dots, K$, with m cells shared across k):

Symbol	Shape	Meaning
$\mathbf{X}_{\text{scATAC}}^{(k)}$	$n_k \times m$	scATAC bin $\times$ cell counts
$\mathbf{X}_{\text{HiC}}^{(k)}$	$l_k \times l_k$ symmetric	Hi-C contact matrix

Factors (Machima internal naming¹):

Symbol	Shape	Sharing	Note
$\mathbf{W}_{\text{RNA}}^{(k)}$	$n_k \times J$	per- k	ATAC-bin $\times$ component basis
$\mathbf{H}_{\text{RNA}}$	$J \times m$	shared	component $\times$ cell loading
$\mathbf{T}^{(k)}$	$l_k \times n_k$	per- k	translator: Hi-C bin $\leftarrow$ ATAC bin
$\mathbf{H}_{\text{Sym}}$	$J \times J$ symmetric	shared	cell-type contact propensity
$\mathbf{U}^{(k)}$	$l_k \times r$	per- k	low-rank left factor of $\mathbf{T}^{(k)}$
$\mathbf{V}^{(k)}$	$n_k \times r$	per- k	low-rank right factor of $\mathbf{T}^{(k)}$

Derived quantities:

$$\mathbf{G}^{(k)} = \mathbf{T}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)} \quad (l_k \times J) \quad (1)$$

$$\mathbf{Y}_{\text{scATAC}}^{(k)} = \mathbf{W}_{\text{RNA}}^{(k)} \mathbf{H}_{\text{RNA}} \quad (n_k \times m) \quad (2)$$

$$\mathbf{Y}_{\text{HiC}}^{(k)} = \mathbf{G}^{(k)} \mathbf{H}_{\text{Sym}} (\mathbf{G}^{(k)})^\top \quad (l_k \times l_k) \quad (3)$$

Notation conventions: $\odot$ is the Hadamard (element-wise) product. Powers applied to a matrix (e.g. $\mathbf{Y}^{\beta-1}$) are element-wise. $\pi_{\text{scATAC}}^{(k)}, \pi_{\text{HiC}}^{(k)}$ are per-chrom modality weights from `.weight()`. L_1^X, L_2^X are sparsity / ridge penalties for factor X . $\rho(\beta)$ is the β -dependent relaxation exponent.

¹The legacy suffix *RNA* on $\mathbf{W}_{\text{RNA}}$ and $\mathbf{H}_{\text{RNA}}$ dates from the original Machima package, where the gene-expression modality drove this side. In Machima2 / AIRE-experiments these factors parameterize the scATAC bin modality $\mathbf{X}_{\text{scATAC}}$.

2 Beta-divergence gradient and the MU template

The element-wise β -divergence and its gradient:

$$d_\beta(x, y) = \begin{cases} \frac{x^\beta + (\beta - 1)y^\beta - \beta x y^{\beta-1}}{\beta(\beta - 1)} & \beta \notin \{0, 1\} \\ x \log(x/y) - x + y & \beta = 1 \text{ (KL)} \\ x/y - \log(x/y) - 1 & \beta = 0 \text{ (IS)} \end{cases} \quad (4)$$

$$\frac{\partial d_\beta}{\partial y} = y^{\beta-1} - x \cdot y^{\beta-2}. \quad (5)$$

For a matrix loss $D_\beta(\mathbf{X}, \mathbf{Y}) = \sum_{ij} d_\beta(X_{ij}, Y_{ij})$ define the residuals

$$\mathbf{M}_{\text{scATAC}}^{(k)} = \mathbf{Y}_{\text{scATAC}}^{(k), \beta-1} - \mathbf{X}_{\text{scATAC}}^{(k)} \odot \mathbf{Y}_{\text{scATAC}}^{(k), \beta-2} \quad (6)$$

$$\mathbf{M}_{\text{HiC}}^{(k)} = \mathbf{Y}_{\text{HiC}}^{(k), \beta-1} - \mathbf{X}_{\text{HiC}}^{(k)} \odot \mathbf{Y}_{\text{HiC}}^{(k), \beta-2}. \quad (7)$$

The standard non-negative MU rule for a factor Θ with positive (denominator) and negative (numerator) gradient parts is

$$\Theta \leftarrow \Theta \odot \left(\frac{\nabla_{\Theta}^-}{\nabla_{\Theta}^+ + (\text{penalties})} \right)^{\rho(\beta)}.$$

For Frobenius ($\beta = 2$) and KL ($\beta = 1$), $\rho \equiv 1$; for Itakura-Saito ($\beta = 0$), $\rho = 1/2$.

3 Update for $\mathbf{H}_{\text{RNA}}$

Only the scATAC reconstruction depends on $\mathbf{H}_{\text{RNA}}$. Sum over chromosomes:

$$\nabla_{H_{\text{RNA}}}^- = \sum_k \pi_{\text{scATAC}}^{(k)} (\mathbf{W}_{\text{RNA}}^{(k)})^\top (\mathbf{X}_{\text{scATAC}}^{(k)} \odot \mathbf{Y}_{\text{scATAC}}^{(k), \beta-2}) \quad (8)$$

$$\nabla_{H_{\text{RNA}}}^+ = \sum_k \pi_{\text{scATAC}}^{(k)} (\mathbf{W}_{\text{RNA}}^{(k)})^\top \mathbf{Y}_{\text{scATAC}}^{(k), \beta-1} + L_1^{H_{\text{RNA}}} + L_2^{H_{\text{RNA}}} \mathbf{H}_{\text{RNA}} \quad (9)$$

$$\mathbf{H}_{\text{RNA}} \leftarrow \mathbf{H}_{\text{RNA}} \odot \left(\nabla_{H_{\text{RNA}}}^- / \nabla_{H_{\text{RNA}}}^+ \right)^{\rho(\beta)} \quad (10)$$

4 Update for $\mathbf{H}_{\text{Sym}}$

Only the Hi-C reconstruction depends on $\mathbf{H}_{\text{Sym}}$. With $\mathbf{X}_{\text{HiC}}^{(k)}$ symmetric, the gradient is also symmetric, so no factor of 2 from chain rule:

$$\nabla_{H_{\text{Sym}}}^- = \sum_k \pi_{\text{HiC}}^{(k)} (\mathbf{G}^{(k)})^\top (\mathbf{X}_{\text{HiC}}^{(k)} \odot \mathbf{Y}_{\text{HiC}}^{(k), \beta-2}) \mathbf{G}^{(k)} \quad (11)$$

$$\nabla_{H_{\text{Sym}}}^+ = \sum_k \pi_{\text{HiC}}^{(k)} (\mathbf{G}^{(k)})^\top \mathbf{Y}_{\text{HiC}}^{(k), \beta-1} \mathbf{G}^{(k)} + L_1^{H_{\text{Sym}}} + L_2^{H_{\text{Sym}}} \mathbf{H}_{\text{Sym}} \quad (12)$$

$$\mathbf{H}_{\text{Sym}} \leftarrow \mathbf{H}_{\text{Sym}} \odot \left(\nabla_{H_{\text{Sym}}}^- / \nabla_{H_{\text{Sym}}}^+ \right)^{\rho(\beta)} \quad (13)$$

$$\mathbf{H}_{\text{Sym}} \leftarrow \frac{1}{2} (\mathbf{H}_{\text{Sym}} + \mathbf{H}_{\text{Sym}}^\top) \quad (\text{symmetrize}) \quad (14)$$

5 Update for $\mathbf{W}_{\text{RNA}}^{(k)}$

$\mathbf{W}_{\text{RNA}}^{(k)}$ enters *both* reconstructions: linearly in $\mathbf{Y}_{\text{scATAC}}^{(k)}$ and quadratically in $\mathbf{Y}_{\text{HiC}}^{(k)}$ via $\mathbf{G}^{(k)} = \mathbf{T}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)}$. The MU update sums the gradient parts from both sides (the chain-rule factor 2 from $\mathbf{W}$ appearing twice in the symmetric Hi-C reconstruction cancels in the ratio):

$$\nabla_{\text{scATAC},k}^- = \pi_{\text{scATAC}}^{(k)} (\mathbf{X}_{\text{scATAC}}^{(k)} \odot \mathbf{Y}_{\text{scATAC}}^{(k),\beta-2}) \mathbf{H}_{\text{RNA}}^\top \quad (15)$$

$$\nabla_{\text{scATAC},k}^+ = \pi_{\text{scATAC}}^{(k)} \mathbf{Y}_{\text{scATAC}}^{(k),\beta-1} \mathbf{H}_{\text{RNA}}^\top \quad (16)$$

$$\nabla_{\text{HiC},k}^- = \pi_{\text{HiC}}^{(k)} (\mathbf{T}^{(k)})^\top (\mathbf{X}_{\text{HiC}}^{(k)} \odot \mathbf{Y}_{\text{HiC}}^{(k),\beta-2}) \mathbf{T}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)} \mathbf{H}_{\text{Sym}} \quad (17)$$

$$\nabla_{\text{HiC},k}^+ = \pi_{\text{HiC}}^{(k)} (\mathbf{T}^{(k)})^\top \mathbf{Y}_{\text{HiC}}^{(k),\beta-1} \mathbf{T}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)} \mathbf{H}_{\text{Sym}} \quad (18)$$

$$\nabla_{W,k}^- = \nabla_{\text{scATAC},k}^- + \nabla_{\text{HiC},k}^- \quad (19)$$

$$\nabla_{W,k}^+ = \nabla_{\text{scATAC},k}^+ + \nabla_{\text{HiC},k}^+ + L_1^{W_{\text{RNA}}} + L_2^{W_{\text{RNA}}} \mathbf{W}_{\text{RNA}}^{(k)} \quad (20)$$

$$\mathbf{W}_{\text{RNA}}^{(k)} \leftarrow \mathbf{W}_{\text{RNA}}^{(k)} \odot \left(\nabla_{W,k}^- / \nabla_{W,k}^+ \right)^{\rho(\beta)} \quad (21)$$

This is where the cross-modality coupling lives: only $\mathbf{W}_{\text{RNA}}^{(k)}$ receives gradient from *both* $\mathbf{X}_{\text{scATAC}}^{(k)}$ and $\mathbf{X}_{\text{HiC}}^{(k)}$.

6 Update for $\mathbf{T}^{(k)}$ (full dense)

Define $\mathbf{A}^{(k)} = \mathbf{W}_{\text{RNA}}^{(k)} \mathbf{H}_{\text{Sym}} (\mathbf{W}_{\text{RNA}}^{(k)})^\top$ ($n_k \times n_k$, symmetric). Then $\mathbf{Y}_{\text{HiC}}^{(k)} = \mathbf{T}^{(k)} \mathbf{A}^{(k)} (\mathbf{T}^{(k)})^\top$ and (factor 2 cancels):

$$\nabla_{T,k}^- = \pi_{\text{HiC}}^{(k)} (\mathbf{X}_{\text{HiC}}^{(k)} \odot \mathbf{Y}_{\text{HiC}}^{(k),\beta-2}) \mathbf{G}^{(k)} \mathbf{H}_{\text{Sym}} (\mathbf{W}_{\text{RNA}}^{(k)})^\top \quad (22)$$

$$\nabla_{T,k}^+ = \pi_{\text{HiC}}^{(k)} \mathbf{Y}_{\text{HiC}}^{(k),\beta-1} \mathbf{G}^{(k)} \mathbf{H}_{\text{Sym}} (\mathbf{W}_{\text{RNA}}^{(k)})^\top + L_1^T + L_2^{T,\text{eff}} \mathbf{T}^{(k)} \quad (23)$$

$$\mathbf{T}^{(k)} \leftarrow \mathbf{T}^{(k)} \odot \left(\nabla_{T,k}^- / \nabla_{T,k}^+ \right)^{\rho(\beta)} \quad (24)$$

where $L_2^{T,\text{eff}} = L_2^T + \lambda_T$ if $T_{\text{regularization}} = 12$, else $L_2^{T,\text{eff}} = L_2^T$.

Frobenius unit rescale. If $T_{\text{regularization}} = \text{frobenius_unit}$, immediately after the MU update:

$$c = \max \left(\sqrt{\frac{1}{K} \sum_k \|\mathbf{T}^{(k)}\|_F^2}, \epsilon \right), \quad \epsilon = \sqrt{\text{.Machine\$double.eps}} \approx 1.49 \times 10^{-8} \quad (25)$$

$$\mathbf{T}^{(k)} \leftarrow \mathbf{T}^{(k)} / c \quad (\forall k) \quad (26)$$

$$\mathbf{H}_{\text{Sym}} \leftarrow c^2 \mathbf{H}_{\text{Sym}} \quad (27)$$

This preserves $\mathbf{Y}_{\text{HiC}}^{(k)}$ exactly because the same scalar c is applied to every $\mathbf{T}^{(k)}$, while $\mathbf{H}_{\text{Sym}}$ is shared.

7 Update for $\mathbf{T}^{(k)} = \mathbf{U}^{(k)}(\mathbf{V}^{(k)})^\top$ (low-rank)

The low-rank parameterization replaces the dense $\mathbf{T}^{(k)}$ with two factors $\mathbf{U}^{(k)} : l_k \times r$, $\mathbf{V}^{(k)} : n_k \times r$. The chain rule gives

$$\frac{\partial L}{\partial \mathbf{U}^{(k)}} = \frac{\partial L}{\partial \mathbf{T}^{(k)}} \mathbf{V}^{(k)}, \quad \frac{\partial L}{\partial \mathbf{V}^{(k)}} = \left(\frac{\partial L}{\partial \mathbf{T}^{(k)}} \right)^\top \mathbf{U}^{(k)}.$$

7.1 $\mathbf{U}^{(k)}$ update

Define a shared right intermediate $\mathbf{B}^{(k)} = \mathbf{H}_{\text{Sym}}(\mathbf{W}_{\text{RNA}}^{(k)})^\top \mathbf{V}^{(k)}$ ($J \times r$). Then:

$$\nabla_{U,k}^- = \pi_{\text{HiC}}^{(k)} (\mathbf{X}_{\text{HiC}}^{(k)} \odot \mathbf{Y}_{\text{HiC}}^{(k), \beta-2}) \mathbf{G}^{(k)} \mathbf{B}^{(k)} \quad (28)$$

$$\nabla_{U,k}^+ = \pi_{\text{HiC}}^{(k)} \mathbf{Y}_{\text{HiC}}^{(k), \beta-1} \mathbf{G}^{(k)} \mathbf{B}^{(k)} + L_1^T + L_2^T \mathbf{U}^{(k)} \quad (29)$$

$$\mathbf{U}^{(k)} \leftarrow \mathbf{U}^{(k)} \odot \left(\nabla_{U,k}^- / \nabla_{U,k}^+ \right)^{\rho(\beta)} \quad (30)$$

7.2 $\mathbf{V}^{(k)}$ update

After the $\mathbf{U}^{(k)}$ update, recompute the materialized quantities $\mathbf{T}^{(k)} \leftarrow \mathbf{U}^{(k)}(\mathbf{V}^{(k)})^\top$, $\mathbf{G}^{(k)} \leftarrow \mathbf{T}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)}$, $\mathbf{Y}_{\text{HiC}}^{(k)} \leftarrow \mathbf{G}^{(k)} \mathbf{H}_{\text{Sym}}(\mathbf{G}^{(k)})^\top$ (Gauss-Seidel ordering for faster convergence than independent updates). Then:

$$\nabla_{V,k}^- = \pi_{\text{HiC}}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)} \mathbf{H}_{\text{Sym}}(\mathbf{G}^{(k)})^\top (\mathbf{X}_{\text{HiC}}^{(k)} \odot \mathbf{Y}_{\text{HiC}}^{(k), \beta-2}) \mathbf{U}^{(k)} \quad (31)$$

$$\nabla_{V,k}^+ = \pi_{\text{HiC}}^{(k)} \mathbf{W}_{\text{RNA}}^{(k)} \mathbf{H}_{\text{Sym}}(\mathbf{G}^{(k)})^\top \mathbf{Y}_{\text{HiC}}^{(k), \beta-1} \mathbf{U}^{(k)} + L_1^T + L_2^T \mathbf{V}^{(k)} \quad (32)$$

$$\mathbf{V}^{(k)} \leftarrow \mathbf{V}^{(k)} \odot \left(\nabla_{V,k}^- / \nabla_{V,k}^+ \right)^{\rho(\beta)} \quad (33)$$

Re-materialize $\mathbf{T}^{(k)} \leftarrow \mathbf{U}^{(k)}(\mathbf{V}^{(k)})^\top$ so the rest of the loop sees a consistent $\mathbf{T}^{(k)}$.

Initialization

- No user-supplied $\mathbf{T}$: $\mathbf{U}^{(k)}, \mathbf{V}^{(k)} \sim \text{Uniform}(0, 1)$ element-wise.
- User-supplied dense $\mathbf{T}^{(k)}$: take rank- r truncated SVD $\mathbf{T}^{(k)} = \mathbf{U}_{\text{svd}} \mathbf{S} \mathbf{V}_{\text{svd}}^\top$, then $\mathbf{U}^{(k)} = |\mathbf{U}_{\text{svd}}| \cdot \sqrt{\mathbf{S}}$, $\mathbf{V}^{(k)} = |\mathbf{V}_{\text{svd}}| \cdot \sqrt{\mathbf{S}}$ (absolute value enforces non-negativity; the symmetric square-root split balances the magnitude).

After initialization, SVD is *not* re-invoked. The main loop only multiplicatively updates $\mathbf{U}, \mathbf{V}$.

8 Algorithm pseudocode

Algorithm 1 Machima2 outer loop (one outer iteration of the symmetric variant)

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1: Input:  $\{\mathbf{X}_{\text{scATAC}}^{(k)}\}, \{\mathbf{X}_{\text{HiC}}^{(k)}\}$ , init for  $\{\mathbf{W}_{\text{RNA}}^{(k)}\}, \mathbf{H}_{\text{RNA}}, \mathbf{H}_{\text{Sym}}$ , and either  $\{\mathbf{T}^{(k)}\}$  or  $\{\mathbf{U}^{(k)}, \mathbf{V}^{(k)}\}$ 
2: Hyperparameters:  $J, \beta, \rho$ , num_iter,  $T_{\text{reg}} \in \{\text{none}, \text{frobenius\_unit}, \text{l2}, \text{low\_rank}\}$ ,  $T_{\text{rank}}$  (if low_rank),  $\lambda_T$  (if l2), L1/L2 penalties, fix flags
3: for iter = 1, ..., num_iter do
4:   if  $\neg \text{fixT}$  then
5:     if  $T_{\text{reg}} = \text{low\_rank}$  then
6:       Update  $\mathbf{U}^{(k)}$  for all  $k$  (Sec. 7.1); rematerialize  $\mathbf{T}^{(k)}, \mathbf{G}^{(k)}, \mathbf{Y}_{\text{HiC}}^{(k)}$ 
7:       Update  $\mathbf{V}^{(k)}$  for all  $k$  (Sec. 7.2); rematerialize  $\mathbf{T}^{(k)}$ 
8:     else
9:       Update  $\mathbf{T}^{(k)}$  for all  $k$  (Sec. 6)
10:      if  $T_{\text{reg}} = \text{frobenius\_unit}$  then
11:         $c \leftarrow \max\left(\sqrt{\frac{1}{K} \sum_k \|\mathbf{T}^{(k)}\|_F^2}, \epsilon\right)$ 
12:         $\mathbf{T}^{(k)} \leftarrow \mathbf{T}^{(k)} / c$  for all  $k$ ;  $\mathbf{H}_{\text{Sym}} \leftarrow c^2 \mathbf{H}_{\text{Sym}}$ 
13:      end if
14:    end if
15:  end if
16:  if  $\neg \text{fixW}_{\text{RNA}}$  then Update  $\mathbf{W}_{\text{RNA}}^{(k)}$  for all  $k$  (Sec. 5)
17:  end if
18:  if  $\neg \text{fixH}_{\text{RNA}}$  then Update  $\mathbf{H}_{\text{RNA}}$  (Sec. 3)
19:  end if
20:  if  $\neg \text{fixH}_{\text{Sym}}$  then Update  $\mathbf{H}_{\text{Sym}}$  and symmetrize (Sec. 4)
21:  end if
22:  Compute reconstruction error and relative change; check convergence threshold
23: end for
24: Return: all factors and (if low_rank)  $\mathbf{U}, \mathbf{V}$  as T_factors

```

Notes on the order of updates. Updating $\mathbf{T}$ (or $\mathbf{U}, \mathbf{V}$) before $\mathbf{W}_{\text{RNA}}$ means the next $\mathbf{W}_{\text{RNA}}$ update sees the freshest reconstruction; updating $\mathbf{H}_{\text{Sym}}$ last, after $\mathbf{T}$ and $\mathbf{W}$ have moved, lets it absorb the residual scale (especially relevant under the `frobenius_unit` rescale). Each step is a strict descent in the multi-modal β -divergence under the standard MU sufficient conditions ($\rho \leq 1$, non-negative inputs).

Cost per outer iteration (low_rank vs full). Dense $\mathbf{T}^{(k)}$ requires $O(l_k^2 n_k)$ work per update; the low-rank parameterization reduces this to $O((l_k + n_k) r J)$ when $r \ll \min(l_k, n_k)$, a reduction of two orders of magnitude at $r = 14$, $J = 7$ on a 1900×1900 block.